Memory (Group 4)

Author roles

Author roles were classified using the Contributor Role Taxonomy (CRediT; credit.niso.org) as follows:

Martina Iscar: project administration, conceptualization, data curation, formal analysis, methodology, investigation, software, validation and writing review and editing.

Irene Morales: project administration, conceptualization, data curation, software, formal analysis, visualization writing review and editing.

Amalia Varo: project administration, conceptualization, data curation, software, formal analysis, formal analysis, methodology, visualization writing review and editing.

Ferrán Extremera: Project administration, conceptualization, data curation, methodology, formal analysis, software, validation visualization writing review and editing.

Use of Generative AI in This Assignment

This project used generative AI as a collaborative tool. Specifically, we employed ChatGPT and Gemini to develop the code by giving it specified prompts, clarify interpretations and improve the storytelling flow.

Even though, these tools were part of the foundation of the code, human intervention has present at each step frequently modifying and correcting overcomplicated code and false interpretations and syntaxis. We understood the databases much more clearly than the AI. Therefore, we were the primary decision makers. We remain responsible for the content, explanations and conclusions drawn. Ai was a tool for enhancing not replace our critical thinking.

Critical Reflection:

Our experience from the first and second case using AI made us more aware of the deficiencies and what strategy works best. Therefore, to take advantage of what we have learnt, we continued with a section approach (each person developed a prompt). However, one thing we did different was to first get AI to explain the overall purpose of the project, what were the prompts' purpose and how they work as a whole to analyse the problem and the objective. This way as a group we all had an overall understanding of the project and we were able to refocus form the start the AI, making it more effective and gaining deeper insights.

Similarly to the first and second project we used AI to give the code. We went one step further encouraging AI to make better graphics and tables. While keeping it simple or our understanding but also taking care of the presentation. This was done by using the knitr library.

In addition, we did not forget to have an active role in the generation of code, making sure the AI did not hallucinate or make assumptions. There were some problems with the approach chosen by the AI to give us results for prompt 3. We had to give continuous feedback and corrections to obtain the results finally presented.

We were careful with the use of AI, because it sometimes tends to overcomplicate the generated text o code. Reinforcement from the human perspective was needed and for creating more concrete prompts for the AI.

As well, we used AI as a tool for interpretations and translator, sometimes concepts or our own prompts are clearer in Spanish and then require the translation of the text, keeping in mind the tone and delivery of the final explanations.

Overall, AI did provide powerful insights and helped us deepen our understanding of the case. Nevertheless, it is our responsibility to proofread the answers given and correct where necessary.

Examples of Significant AI Prompt–Response Interactions

Example 1:

> **Prompt:** With the dataset provided, conduct a Kaplan–Meier survival analysis to evaluate whether there are differences in employee turnover over tenure by gender. Compare survival curves, interpret the results, and explain at which stages of tenure these differences are most pronounced.
>
> **Response:** It generated the R code required to perform the Kaplan–Meier survival analysis, including the creation of the survival object, estimation of survival curves by gender, and the statistical comparison using the log-rank test. It also provided guidance on how to visualise the results and interpret both the survival curves and the statistical output, highlighting when differences between groups become more evident over time.
>
> **Evaluation:** We reviewed the results obtained from the analysis and compared them with our own understanding of survival analysis concepts. We confirmed that the interpretation was consistent with the graphical output, particularly regarding the stages of tenure where differences between genders become more pronounced. This allowed us to better understand the turnover patterns and validate the conclusions drawn from the model.

Example 2:

Prompt: With the source uploaded help, me develop a Logistic regression output and table of log-odds and odds ratios. Explain every line of code and compere the logistic regression with another model to see if it is good enough.

Response: It gave the code of the prompt 3 and explained every line. In addition, it comperes the logistic regression model with a null model to verify that the logistic model is better in this case. It asked if we wanted to analyse the solution and if we wanted recommendations for the company.

Evaluation: The code wasn't helpful enough as every time we tried to run it was giving problems. After few more prompts, we were able to get the code without problems. The explanation of the code was helpful; it allowed us to understand what we were writing in R. However, we must be careful because if we weren't able to understand the code and find the mistake, we might have loss more time than required.

Example 3:

Prompt: Taking into account the images uploaded and the information I have given you analyse the validity of the test and how it can be solved or improved. Mention and integrate into the response Fisher test and Monte Carlo. Also from an HR perspective evaluate if this is a concern or not

- **Response**: It explicitly tells us that HR should not launch new retention programs or panic based on this single test. It arguments that the math does not strongly prove that the country an employee lives in is the main reason they are quitting. Also explained the technical glitch in the test (the Chi-square warning). It interpreted the p_value and gave attention to the two outliers commenting on them.

Evaluation: After reflecting on the explanation, we carefully went over it making sure with our knowledge form statistics and previous cases that the answers were accurate. Also, we took into account the R-squared and evaluated if it is high or low and why the ai said it is high, even though it is about 20%, but from what we learned in class and was commented we concluded that it is high due to the type of data and analysis. To get this level of explainability is considerable.

Example 4:

Prompt: Give an overall explanation of the objective of the analysis and how each prompt contributes to the analysis

Response: The main goal of this analysis is to move beyond basic HR reporting and use advanced statistics to truly understand why employees quit, allowing the company to make data-driven decisions.

Here is how each prompt builds the story:

Prompts 1 & 2 (The "Where"): helping HR prioritize which regions need immediate attention.

Prompt 3 (The "Who"): This looks at individual traits to predict exactly which types of employees are most at risk of resigning.

Prompt 4 (The "When"): This tracks the timeline of employees' careers to pinpoint at what specific stage of tenure men and women are most likely to leave the company.

Prompt 5 (The "Why"): This evaluates team culture and environment to figure out what drives turnover.

Prompt 6 (The "Cost"): translates all the statistical findings into a financial business case.

Evaluation: We used this information to better understand as a group how to target each prompt and what to expect and the logic behind it. Allowing us to have critical thinking immediately when asking AI to develop the code, because we had an overall view of everything